

Class 19: Pertussis Mini Project

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Background

Pertussis is a bacterial lung infection also known as Whooping cough. Let's begin by examining CDC reported case numbers in the US.

```
cdc <- data.frame(
  Year = c(1922L,
           1923L, 1924L, 1925L, 1926L, 1927L, 1928L,
           1929L, 1930L, 1931L, 1932L, 1933L, 1934L, 1935L,
           1936L, 1937L, 1938L, 1939L, 1940L, 1941L,
           1942L, 1943L, 1944L, 1945L, 1946L, 1947L, 1948L,
           1949L, 1950L, 1951L, 1952L, 1953L, 1954L,
           1955L, 1956L, 1957L, 1958L, 1959L, 1960L,
           1961L, 1962L, 1963L, 1964L, 1965L, 1966L, 1967L,
           1968L, 1969L, 1970L, 1971L, 1972L, 1973L,
           1974L, 1975L, 1976L, 1977L, 1978L, 1979L, 1980L,
           1981L, 1982L, 1983L, 1984L, 1985L, 1986L,
           1987L, 1988L, 1989L, 1990L, 1991L, 1992L, 1993L,
           1994L, 1995L, 1996L, 1997L, 1998L, 1999L,
           2000L, 2001L, 2002L, 2003L, 2004L, 2005L,
           2006L, 2007L, 2008L, 2009L, 2010L, 2011L, 2012L,
           2013L, 2014L, 2015L, 2016L, 2017L, 2018L,
           2019L, 2020L, 2021L, 2022L, 2023L, 2024L),
  No..Reported.Pertussis.Cases = c(107473,
                                   164191, 165418, 152003, 202210, 181411,
```

```

161799,197371,166914,172559,215343,179135,
265269,180518,147237,214652,227319,103188,
183866,222202,191383,191890,109873,
133792,109860,156517,74715,69479,120718,
68687,45030,37129,60886,62786,31732,28295,
32148,40005,14809,11468,17749,17135,
13005,6799,7717,9718,4810,3285,4249,
3036,3287,1759,2402,1738,1010,2177,2063,
1623,1730,1248,1895,2463,2276,3589,
4195,2823,3450,4157,4570,2719,4083,6586,
4617,5137,7796,6564,7405,7298,7867,
7580,9771,11647,25827,25616,15632,10454,
13278,16858,27550,18719,48277,28639,
32971,20762,17972,18975,15609,18617,6124,
2116,3044,7063,22538)
)
head(cdc)

```

	Year	No..Reported.Pertussis.Cases
1	1922	107473
2	1923	164191
3	1924	165418
4	1925	152003
5	1926	202210
6	1927	181411

Plot of cases per year for Pertussis in the US

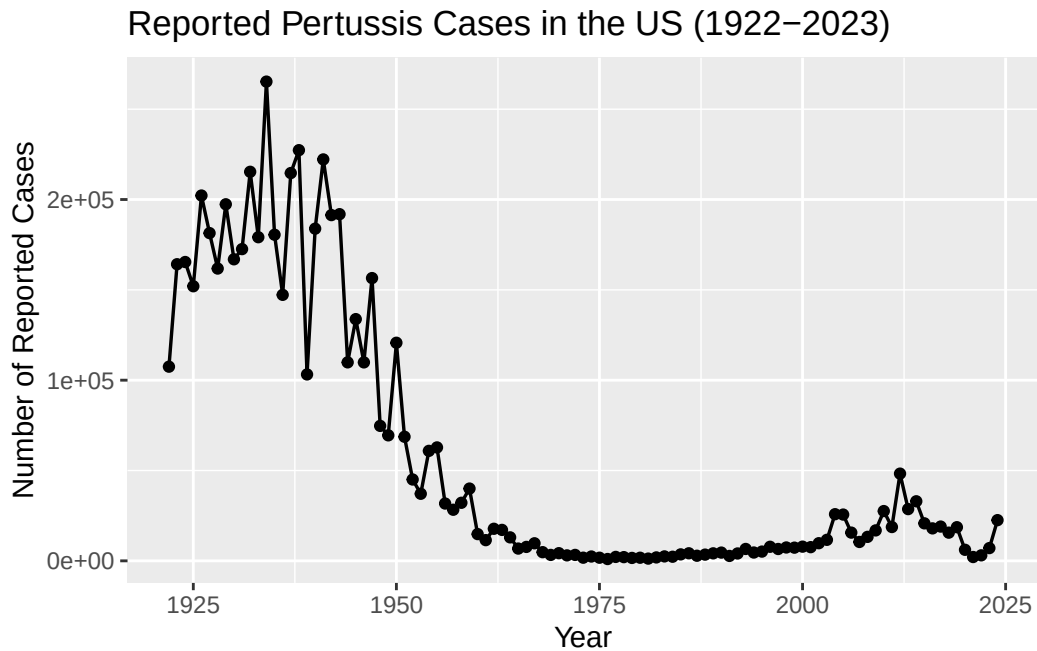
```

library(ggplot2)

ggplot(cdc, aes(x = Year, y = No..Reported.Pertussis.Cases)) +
  geom_line(size = .60) +
  geom_point() +
  labs(title = "Reported Pertussis Cases in the US (1922-2023)",
       x = "Year",
       y = "Number of Reported Cases")

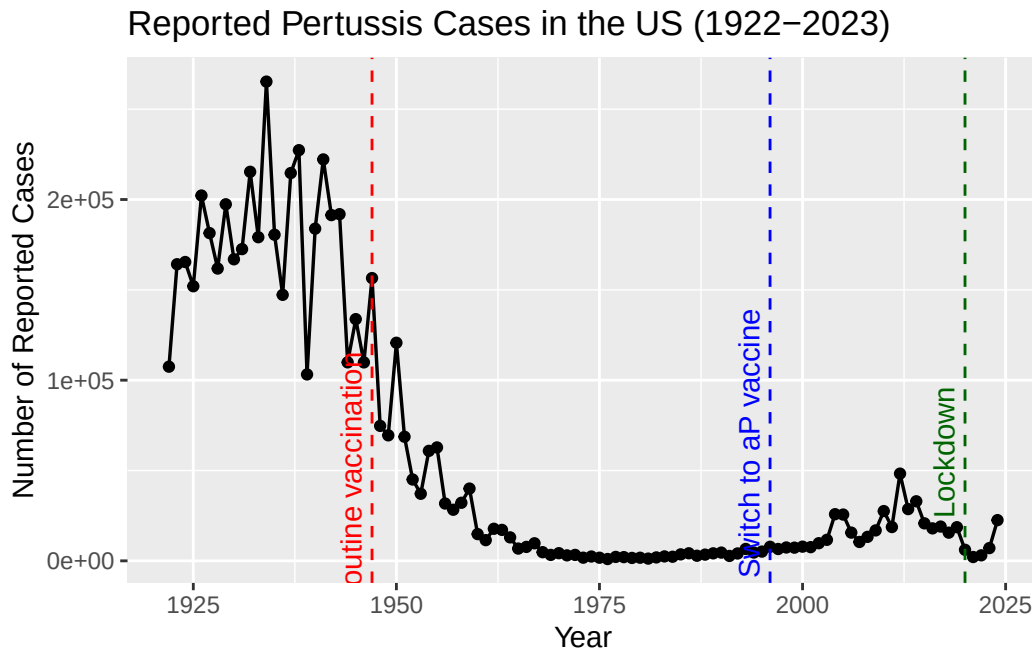
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 i Please use `linewidth` instead.



Add some major milestone timepoints into our plot:

```
ggplot(cdc, aes(x = Year, y = No..Reported.Pertussis.Cases)) +
  geom_line(size = .60) +
  geom_point() +
  labs(title = "Reported Pertussis Cases in the US (1922-2023)",
       x = "Year",
       y = "Number of Reported Cases") +
  geom_vline(xintercept = 1947, linetype="dashed", color = "red") +
  geom_vline(xintercept = 1996, linetype="dashed", color = "blue") +
  geom_vline(xintercept = 2020, linetype="dashed", color = "darkgreen") +
  annotate("text", x = 1947, y = 45000, label = "Routine vaccination", angle=90, vjust=-0.5,
  annotate("text", x = 1996, y = 65000, label = "Switch to aP vaccine", angle=90, vjust=-0.5,
  annotate("text", x = 2020, y = 60000, label = "Lockdown", angle=90, vjust=-0.5, color="darkgreen")
```



The full introduction of mandatory wP (whole-cell) Pertussis immunization in the mid 1940s lead to a dramatic reduction in case numbers (from over 200,000 to 100s).

The switch to the aP (newer acellular) formalization precedes a gradual increase in case numbers over the next two decades, peaking in 2012 with over 48,000 reported cases.

Finally, the recent COVID-19 lockdowns and social distancing measures appear to have led to a sharp drop in Pertussis cases in 2020 and 2021, with a slight rebound in 2024.

The CMI-PB project

The mission of the CMI-PB project is to develop a next-generation Pertussis vaccine that provides longer-lasting immunity and better protection against circulating strains.

They make their data available via JSON format API endpoints- basically the database tables in a key:value format. To read this we can use the `jsonlite` package.

```
library(jsonlite)
```

Warning: package 'jsonlite' was built under R version 4.5.2

```
subject <- read_json("https://www.cmi-pb.org/api/v5_1/subject", simplifyVector = TRUE)
head(subject)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White
4	4	wP	Male	Not Hispanic or Latino	Asian
5	5	wP	Male	Not Hispanic or Latino	Asian
6	6	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset
4	1988-01-01	2016-08-29	2020_dataset
5	1991-01-01	2016-08-29	2020_dataset
6	1988-01-01	2016-10-10	2020_dataset

Q. How many “subjects”/individuals are in this dataset?

```
nrow(subject)
```

```
[1] 172
```

Q. How many wP and aP subjects are there?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q. What is the breakdown by “biological_sex” and “race”?

```
table(subject$biological_sex)
```

```
Female    Male
  112      60
```

```
table(subject$race)
```

American Indian/Alaska Native	1
Asian	44
Black or African American	5
More Than One Race	19
Native Hawaiian or Other Pacific Islander	2
Unknown or Not Reported	21
White	80

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

This breakdown is not particularly representative of the US population - this is a serious caveat when interpreting the results of this study. However, it is still the largest sample of it's type every assembled.

```
specimen <- read_json("https://www.cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)
ab_titer <- read_json("https://www.cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
head(specimen)
```

	specimen_id	subject_id	actual_day_relative_to_boost
1	1	1	-3
2	2	1	1
3	3	1	3
4	4	1	7

5	5	1		11
6	6	1		32
	planned_day_relative_to_boost	specimen_type	visit	
1		0	Blood	1
2		1	Blood	2
3		3	Blood	3
4		7	Blood	4
5		14	Blood	5
6		30	Blood	6

```
head(ab_titer)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection
1	UG/ML	2.096133
2	IU/ML	29.170000
3	IU/ML	0.530000
4	IU/ML	6.205949
5	IU/ML	4.679535
6	IU/ML	2.816431

We need to “join” or link these tables with the `subject` table so we can begin to analyze this data and know who a given Ab sample was collected for and when.

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
meta <- inner_join(subject, specimen)
```

Joining with `by = join\_by(subject\_id)`

```
head(meta)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female Not Hispanic or Latino	White	
2	1	wP	Female Not Hispanic or Latino	White	
3	1	wP	Female Not Hispanic or Latino	White	
4	1	wP	Female Not Hispanic or Latino	White	
5	1	wP	Female Not Hispanic or Latino	White	
6	1	wP	Female Not Hispanic or Latino	White	

	year_of_birth	date_of_boost	dataset	specimen_id
1	1986-01-01	2016-09-12	2020_dataset	1
2	1986-01-01	2016-09-12	2020_dataset	2
3	1986-01-01	2016-09-12	2020_dataset	3
4	1986-01-01	2016-09-12	2020_dataset	4
5	1986-01-01	2016-09-12	2020_dataset	5
6	1986-01-01	2016-09-12	2020_dataset	6

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3	0	Blood
2	1	1	Blood
3	3	3	Blood
4	7	7	Blood
5	11	14	Blood
6	32	30	Blood

	visit
1	1
2	2
3	3
4	4
5	5
6	6

Now let's join the `ab_titer` table to `meta` to get the actual Ab measurements.

```
ab_data <- inner_join(meta, ab_titer)
```

Joining with `by = join\_by(specimen\_id)`


```
head(ab_data)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female Not Hispanic or Latino	White	
2	1	wP	Female Not Hispanic or Latino	White	
3	1	wP	Female Not Hispanic or Latino	White	
4	1	wP	Female Not Hispanic or Latino	White	
5	1	wP	Female Not Hispanic or Latino	White	
6	1	wP	Female Not Hispanic or Latino	White	

	year_of_birth	date_of_boost	dataset	specimen_id
1	1986-01-01	2016-09-12	2020_dataset	1
2	1986-01-01	2016-09-12	2020_dataset	1
3	1986-01-01	2016-09-12	2020_dataset	1
4	1986-01-01	2016-09-12	2020_dataset	1
5	1986-01-01	2016-09-12	2020_dataset	1
6	1986-01-01	2016-09-12	2020_dataset	1

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3	0	Blood
2	-3	0	Blood
3	-3	0	Blood
4	-3	0	Blood
5	-3	0	Blood
6	-3	0	Blood

	visit	isotype	is_antigen_specific	antigen	MFI	MFI_normalised	unit
1	1	IgE	FALSE	Total	1110.21154	2.493425	UG/ML
2	1	IgE	FALSE	Total	2708.91616	2.493425	IU/ML
3	1	IgG	TRUE	PT	68.56614	3.736992	IU/ML
4	1	IgG	TRUE	PRN	332.12718	2.602350	IU/ML
5	1	IgG	TRUE	FHA	1887.12263	34.050956	IU/ML
6	1	IgE	TRUE	ACT	0.10000	1.000000	IU/ML

	lower_limit_of_detection
1	2.096133
2	29.170000
3	0.530000
4	6.205949
5	4.679535
6	2.816431

Q. How many Ab measurements are there in total?

```
nrow(ab_data)
```

```
[1] 61956
```

Q. How many different isotypes are there in the `ab_data` table?

```
unique(ab_data$isotype)
```

```
[1] "IgE" "IgG" "IgG1" "IgG2" "IgG3" "IgG4"
```

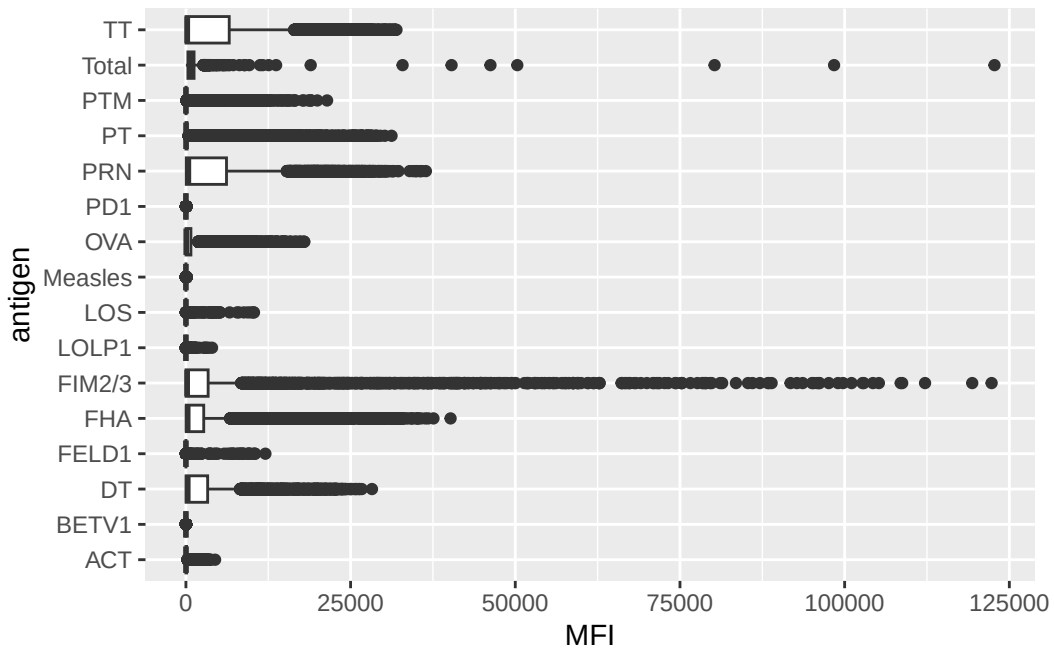
Q. How many different antigens are there in the `ab_data` table?

```
unique(ab_data$antigen)
```

```
[1] "Total" "PT" "PRN" "FHA" "ACT" "LOS" "FELD1"  
[8] "BETV1" "LOLP1" "Measles" "PTM" "FIM2/3" "TT" "DT"  
[15] "OVA" "PD1"
```

```
ggplot(ab_data)+  
  aes(MFI, antigen)+  
  geom_boxplot()
```

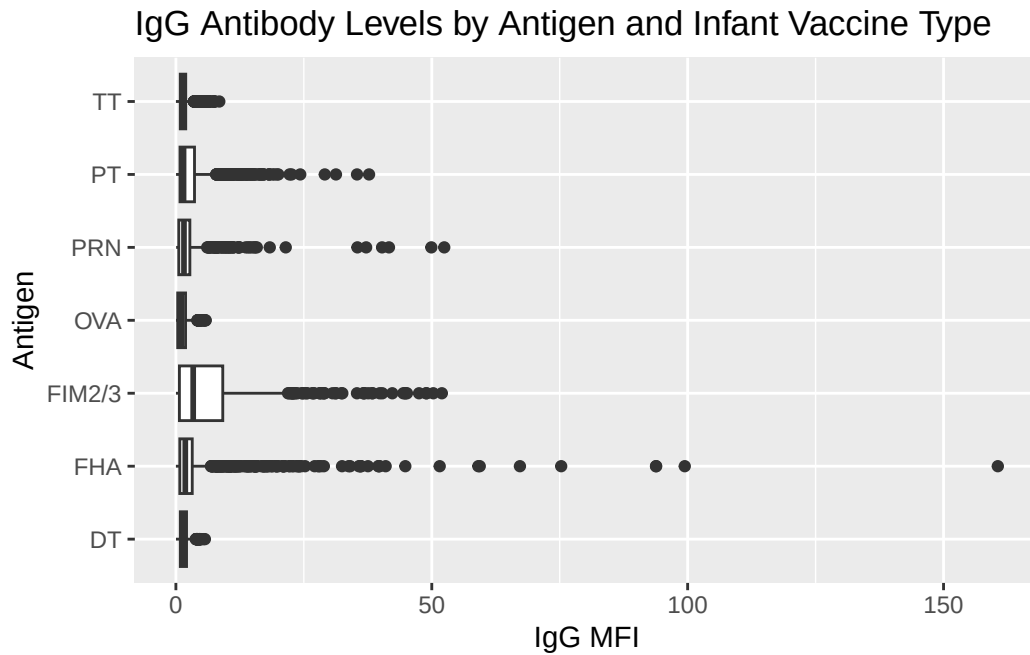
Warning: Removed 1 row containing non-finite outside the scale range (``stat_boxplot()``).



IgG is crucial for long-term immunity against Pertussis, so let's filter to just IgG measurements.

```
ab_data |>
  filter(isotype == "IgG") -> igg
```

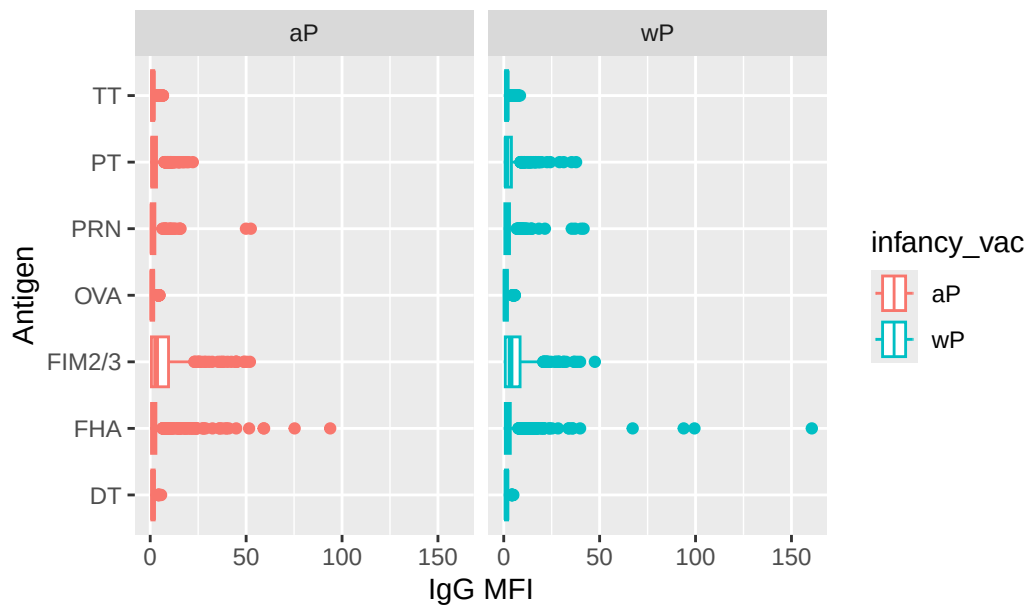
```
ggplot(igg)+
  aes(MFI_normalised, antigen)+
  geom_boxplot(position=position_dodge(width=0.8))+
  labs(title="IgG Antibody Levels by Antigen and Infant Vaccine Type",
        x="IgG MFI",
        y="Antigen")
```



We can “facet” our plot by wP vs aP infant vaccine type to see if there are any differences.

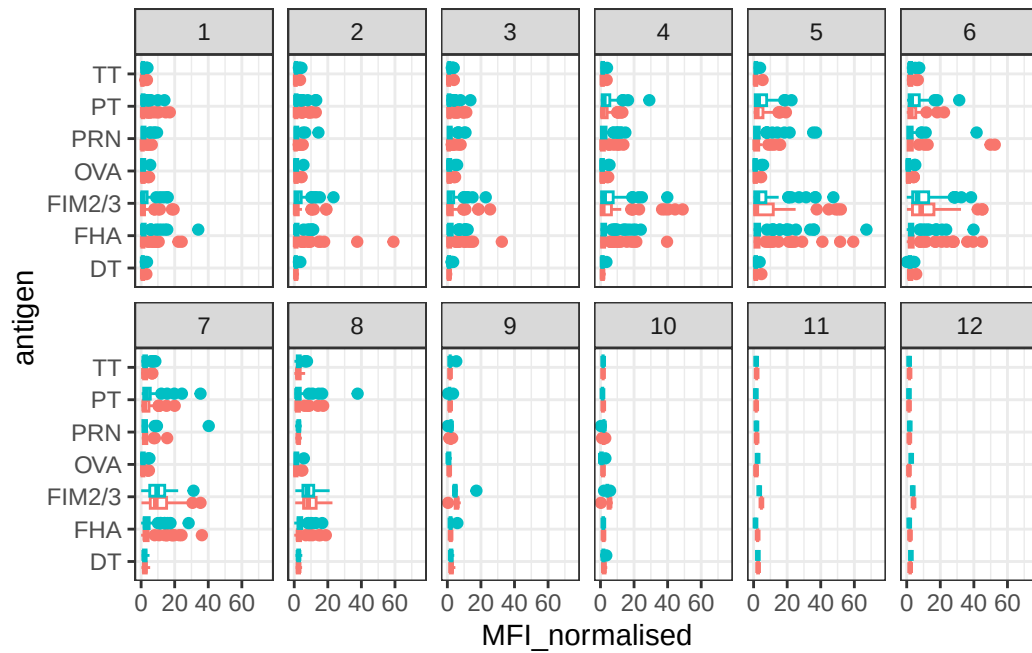
```
ggplot(igg)+
  aes(MFI_normalised, antigen, colour = infancy_vac)+
  geom_boxplot(position=position_dodge(width=0.8))+
  labs(title="IgG Antibody Levels by Antigen and Infant Vaccine Type",
        x="IgG MFI",
        y="Antigen") +
  facet_wrap(~infancy_vac)
```

IgG Antibody Levels by Antigen and Infant Vaccine Type



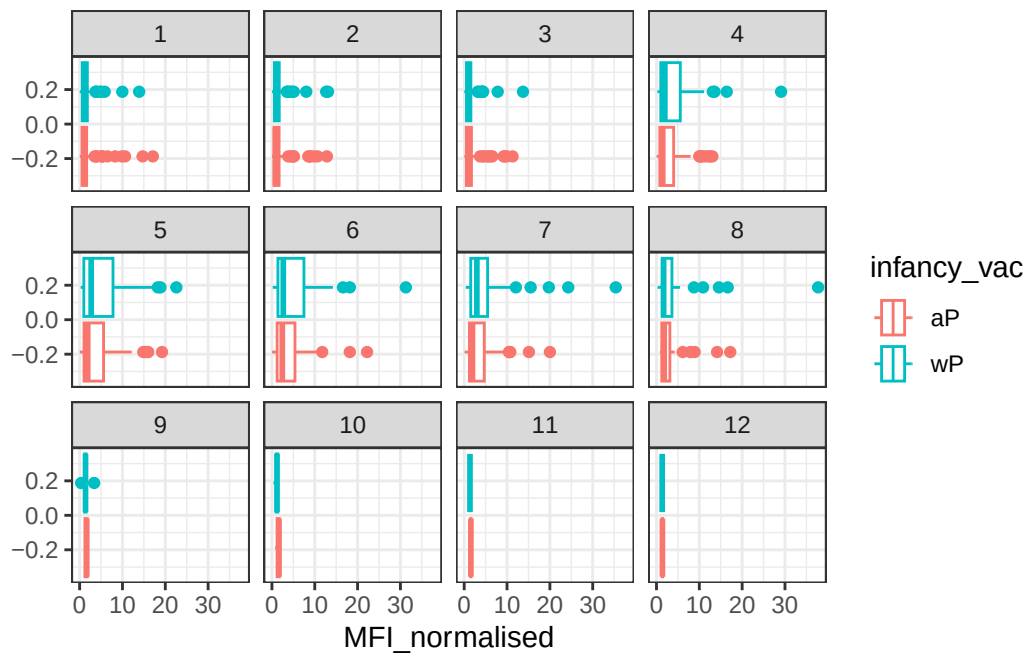
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



More advanced analysis digging into individual antigen responses over time:

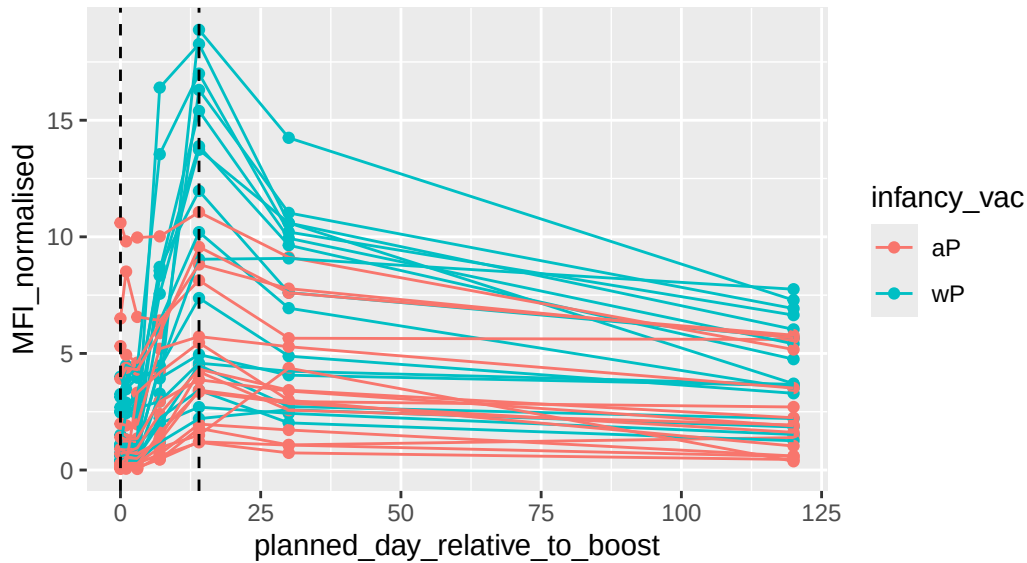
```
filter(igg, antigen=="PT") |>
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = T) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



```
filter(igg, antigen == "PT", dataset == "2021_dataset") |>
ggplot() +
  aes(x=planned_day_relative_to_boost,
      y=MFI_normalised,
      col=infancy_vac,
      group=subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept=0, linetype="dashed") +
  geom_vline(xintercept=14, linetype="dashed") +
  labs(title="2021 dataset IgG PT",
       subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

2021 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)



This graph shows the IgG response to Pertussis Toxin (PT) over time in individuals who received either wP or aP infant vaccines and were boosted with Tdap as adults. The peak in 14 days post-boost happens due to the rapid activation of memory B cells generated from prior vaccination.

There are lots of cool things to explore in this dataset and we need coding and biology knowledge to do it effectively - i.e. you guys!!!